PS200C: Regression Discontinuity Designs

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Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
   - cluster randomization
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
   - sensitivity
4. Difference in Differences
5. Instrumental Variables
6. Regression Discontinuity: Sharp, Fuzzy

Regression Discontinuity Design (RDD)

- RDD is a fairly old idea (Thistlethwaite and Campbell, 1960) but this design experienced a renaissance in recent years.
- Assignment to treatment and control is not random, but:
  - there is a rule influencing how people are assigned
  - at some boundary, a trivial change in expected potential outcome corresponds to a big change in probability of treatment
  - effect estimated is hopefully dominated by effect with only small role for bias.
- Widely applicable in a rule based world (administrative programs, elections, etc.)
- High internal validity, poor external validity

Sharp Regression Discontinuity Design

- Imagine a binary treatment D_i that is **completely** determined by the value of a predictor X_i being on either side of a fixed cutoff point c:

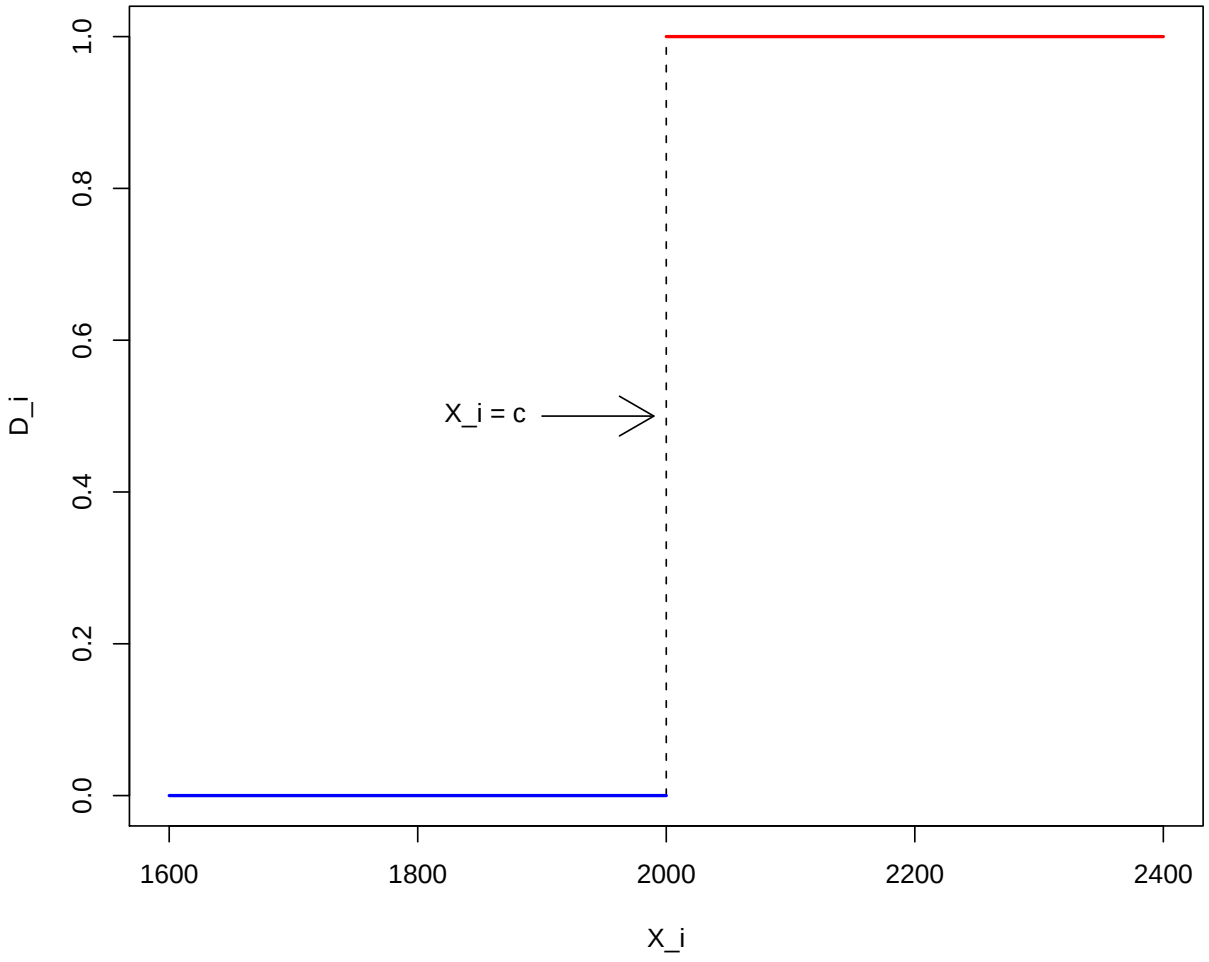
$$D_i = \mathbb{1}_{\{X_i > c\}} \text{ so } D_i = \begin{cases} D_i = 1 & \text{if } X_i > c \\ D_i = 0 & \text{if } X_i < c \end{cases}$$

- Later we will consider treatment D_i that is only probabilistically determined by the cutoff in X_i, which we will call *fuzzy* RDD.
- X_i, called the forcing or running variable, may be correlated with potential outcomes, so comparing treated and untreated units does not provide causal estimates
- Situation arises often from administrative decisions, where policy decisions are made using sharp rules rather than administrator discretion

Hypothetical Example

- Thistlethwaite and Campbell (1960) study the effects of college scholarships on students' later achievements
- Scholarships are given on the basis of whether or not a student's test score exceeds some threshold c
  - Treatment D_i is scholarship
  - *Forcing* or *running* variable X_i is SAT score with cutoff c
  - Outcome Y_i is subsequent earnings; potential outcomes Y_{0i}, Y_{1i}.
- Y_{1i} and Y_{0i} are correlated with X_i (and D_i): on average, students with higher SAT scores obtain higher earnings

Sharp RDD: Graphical Illustration



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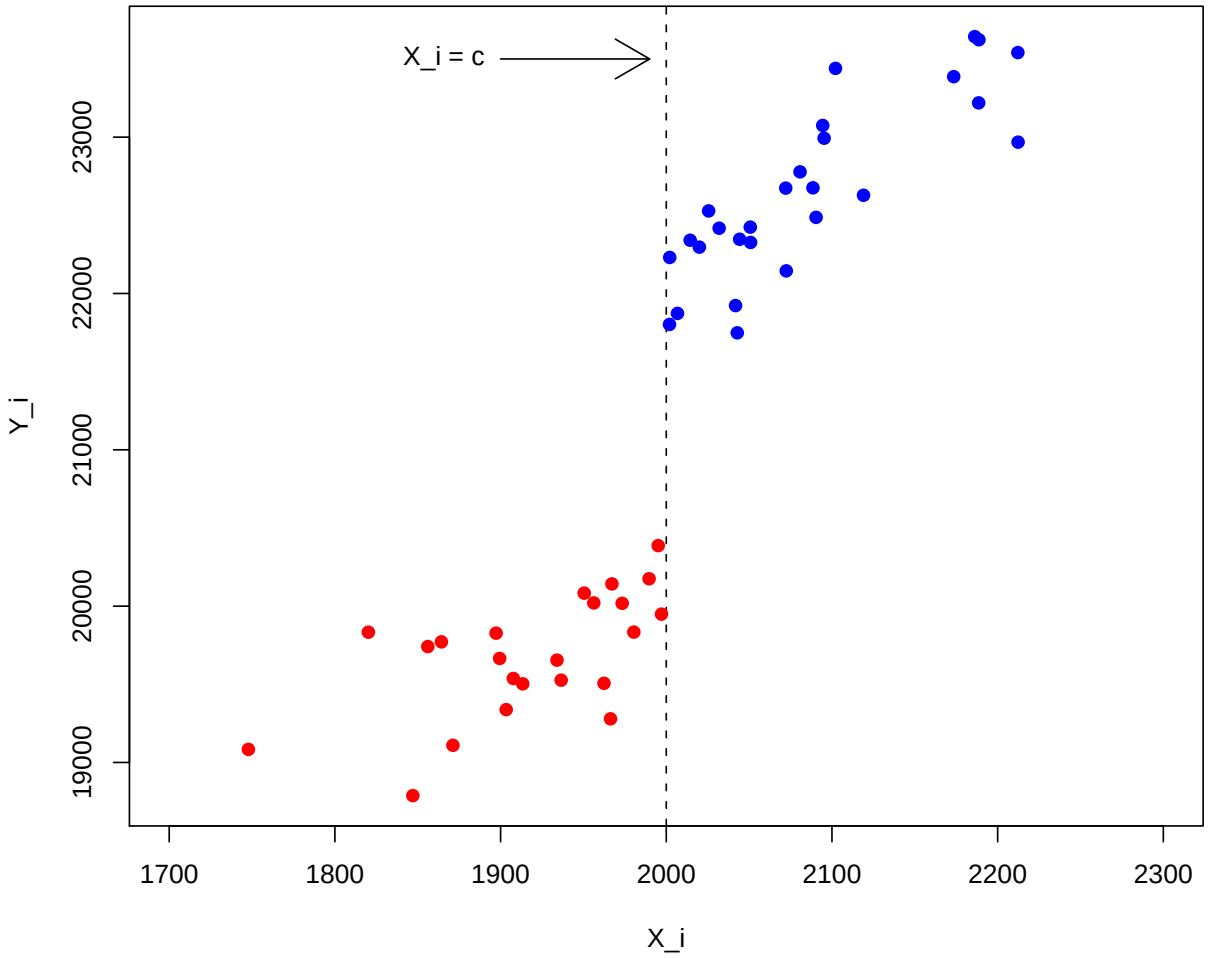

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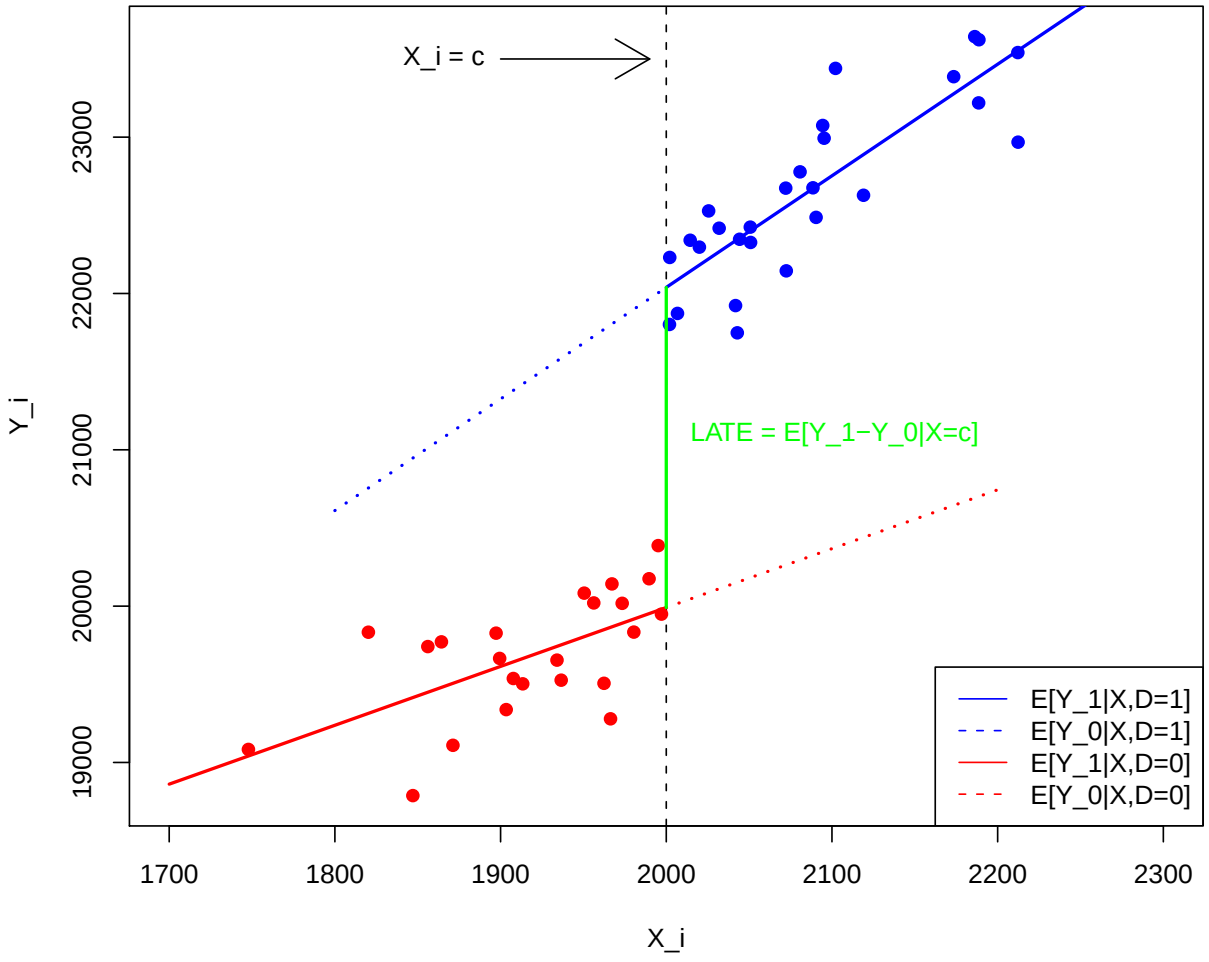


Sharp RDD: Identification

Identification Assumption

$\mathbb{E}[Y_{1i}|X_i]$ *and* $\mathbb{E}[Y_{0i}|X_i]$ *are continuous in* X_i *around the threshold* $X_i = c$

Identification Result

The treatment effect is identified at the threshold as:

$$\begin{aligned}
\alpha_{SRDD} &= \mathbb{E}[Y_{1i} - Y_{0i}|X_i = c] \\
&= \mathbb{E}[Y_{1i}|X_i = c] - \mathbb{E}[Y_{i0}|X_i = c] \\
&= \lim_{x \downarrow c} \mathbb{E}[Y_{1i}|X_i = c] - \lim_{x \uparrow c} \mathbb{E}[Y_{i0}|X_i = c]
\end{aligned}$$

Without further assumptions α_{SRDD} *is only valid at the threshold.*

Comments on Identification

The continuity assumption says a lot:

- By saying $\mathbb{E}[Y_{1i}|X_i]$ and $\mathbb{E}[Y_{0i}|X_i]$ are continuous in X near c, we mean they are "smooth", or do not "jump" at $X = c$.
- This is why we can say:

$$\mathbb{E}[Y_{1i}|X_i = c] = \lim_{\epsilon \downarrow 0} \mathbb{E}[Y_{i1}|X_i = c + \epsilon, D_i = 1]$$
$$\mathbb{E}[Y_{0i}|X_i = c] = \lim_{\epsilon \uparrow 0} \mathbb{E}[Y_{io}|X_i = c + \epsilon, D_i = 0]$$

  where we estimate (rather than observe) each expectation

Allowing us to identify:

$$\hat{\alpha} = \hat{\mathbb{E}}[Y_{1i}|X_i = c] - \hat{\mathbb{E}}[Y_{0i}|X_i = c]$$

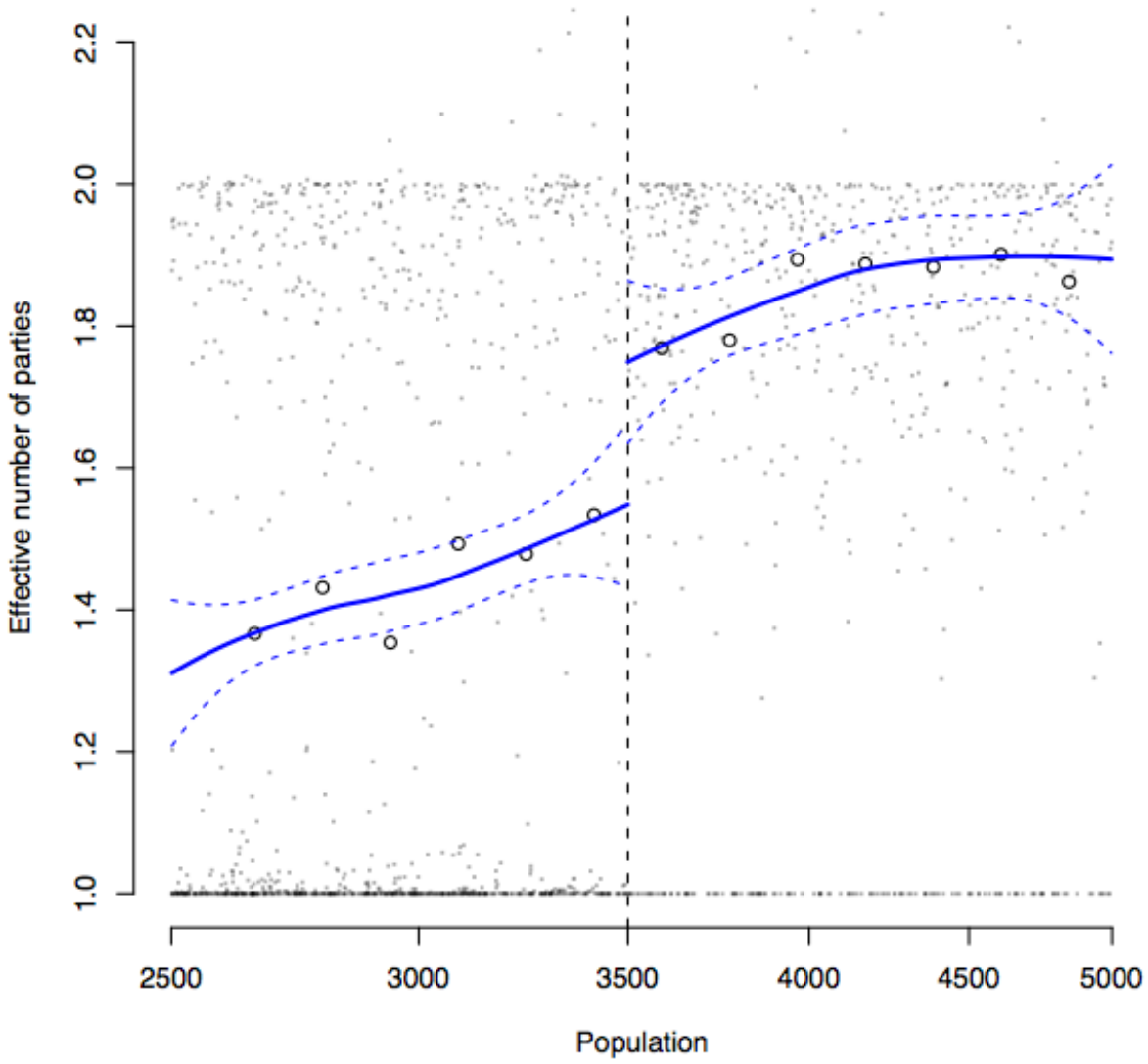
- Implies any *unobserved influences on the potential outcomes* are also continuous in X (at $X = c$)
- Heuristically we sometimes think about "local randomization" implied by this:

$$\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i | X_i \approx c$$

Duverger's Law (Eggers, 2010)

- Duverger (1972): "a majority vote on one ballot is conducive to a two-party system; proportional representation is conducive to a multiparty system"
- Therefore, we expect the number of parties to increase when going from a majority to a proportional electoral system
- In French municipalities, the electoral rule used to elect the municipal council depends on the city's population:
  - cities with fewer than 3,500 people use first-past-post
  - cities with a population of 3,500 or more use a form of PR rule

Sharp RDD: Duverger's Law



Sharp RDD: Electronic Voting (Hidalgo, 2012)

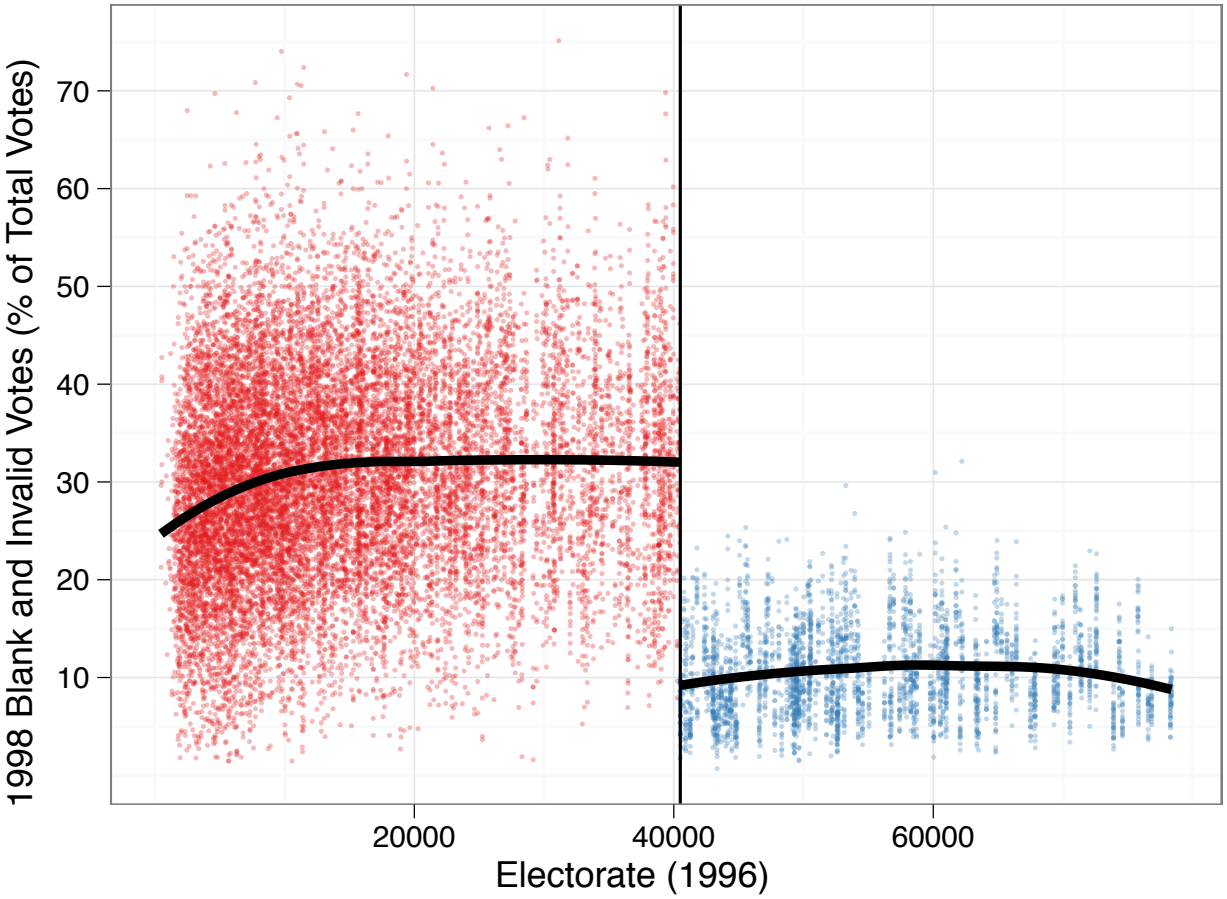
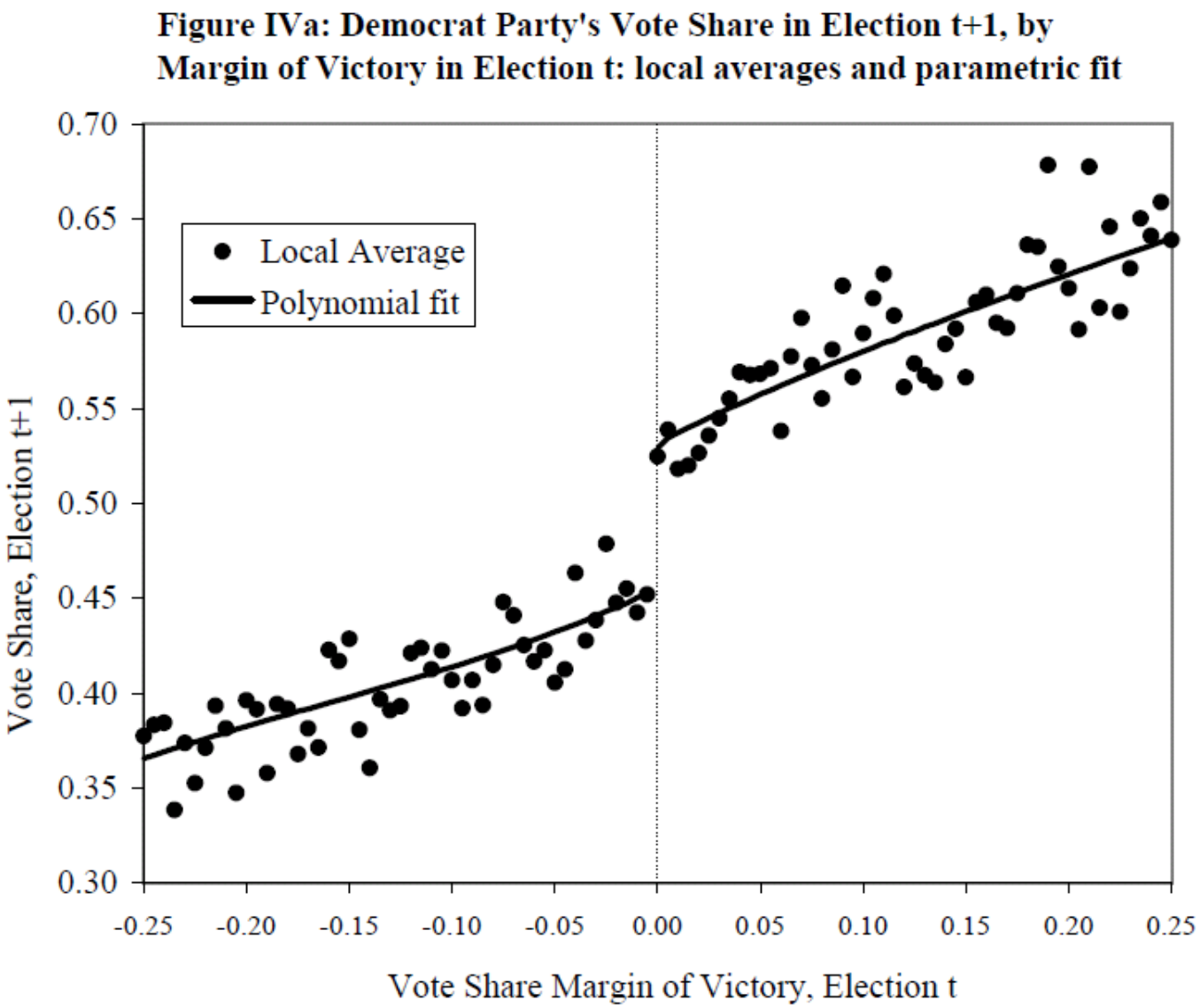


Figure 6: The effect of electronic voting on the percent of null and blank votes. Each dot is a polling station. Polling stations to the left of the vertical black line used paper ballots and polling stations to the right used electronic voting. The black horizontal line is the conditional mean of the outcome estimated with a loess regression.

Sharp RDD: Incumbency Advantage



A Few Other Examples

- Effect of class size on student achievement (Angrist and Lavy 1999)
- Effect of access to credit on development outcomes (loan offer is determined by credit score threshold)
- Effect of wage increase on performance of mayors (Ferraz and Finan 2011, Gagliarducci and Nannicini 2013)
- Effect of wining office on wealth of politicians (Eggers and Hainmueller 2009)
- Effect of school district boundaries on home values (Black 1999)
- RD that exploits “close" elections has become a workhorse model for causal inference in electoral research

  ...Lee, Moretti and Butler 2004, DiNardo and Lee 2004, Hainmueller and Kern 2008, Leigh 2008, Pettersson-Lidbom 2008, Broockman 2009, Butler 2009, Dal Bó, Dal Bó and Snyder 2009, Eggers and Hainmueller 2009, Ferreira and Gyourko 2009, Uppal 2009, 2010, Cellini, Ferreira and Rothstein 2010, Gerber and Hopkins 2011, Trounstine 2011, Boas and Hidalgo 2011, Folke and Snyder Jr. 2012, and Gagliarducci and Paserman 2012...

- But beware of “close election” designs with treatment other than “winning”
  - e.g. gender, party, education
  - RDD does NOT tell us “the effect of that characteristics”
  - it does tell us “the effect of electing somebody with that characteristic” – think about districts, not individuals as the units.

How to actually estimate these?

Estimation preferences for RD may be changing, but here is an overview.

Consider the following options:

- Do something simple: separate, linear or quadratic models.
- Use (separate) local polynomials with some kind of optimality claim for the bandwidth selection and perhaps the degree.
- Use (separate) Gaussian Processes

1. Simple, separate, linear or quadratic models

Suppose $\mathbb{E}[Y_0|X]$ and $\mathbb{E}[Y_1|X]$ are separate polynomial functions of X and the average effect of the treatment $\mathbb{E}[Y_1 - Y_0|X]$ varies with X

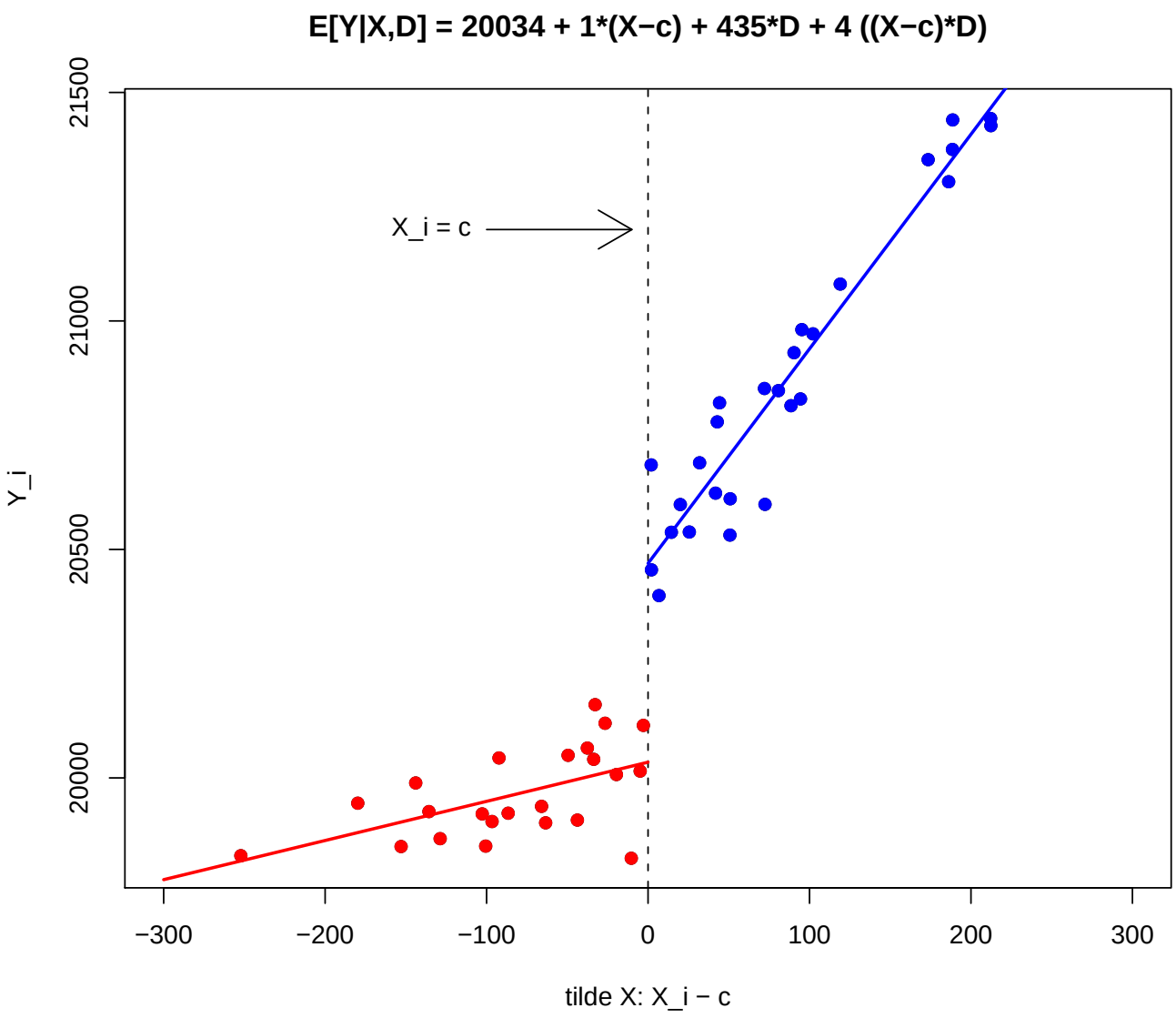
E.g. quadratic,

$$\begin{aligned}\mathbb{E}[Y_i|X_i, D_i] =& \gamma_0 + \gamma_1 \tilde{X}_i + \gamma_2 \tilde{X}_i^2 \\ &+ \alpha_0 D_i + \alpha_1(\tilde{X}_i \cdot D_i) + \alpha_2(\tilde{X}_i^2 \cdot D_i)\end{aligned}$$

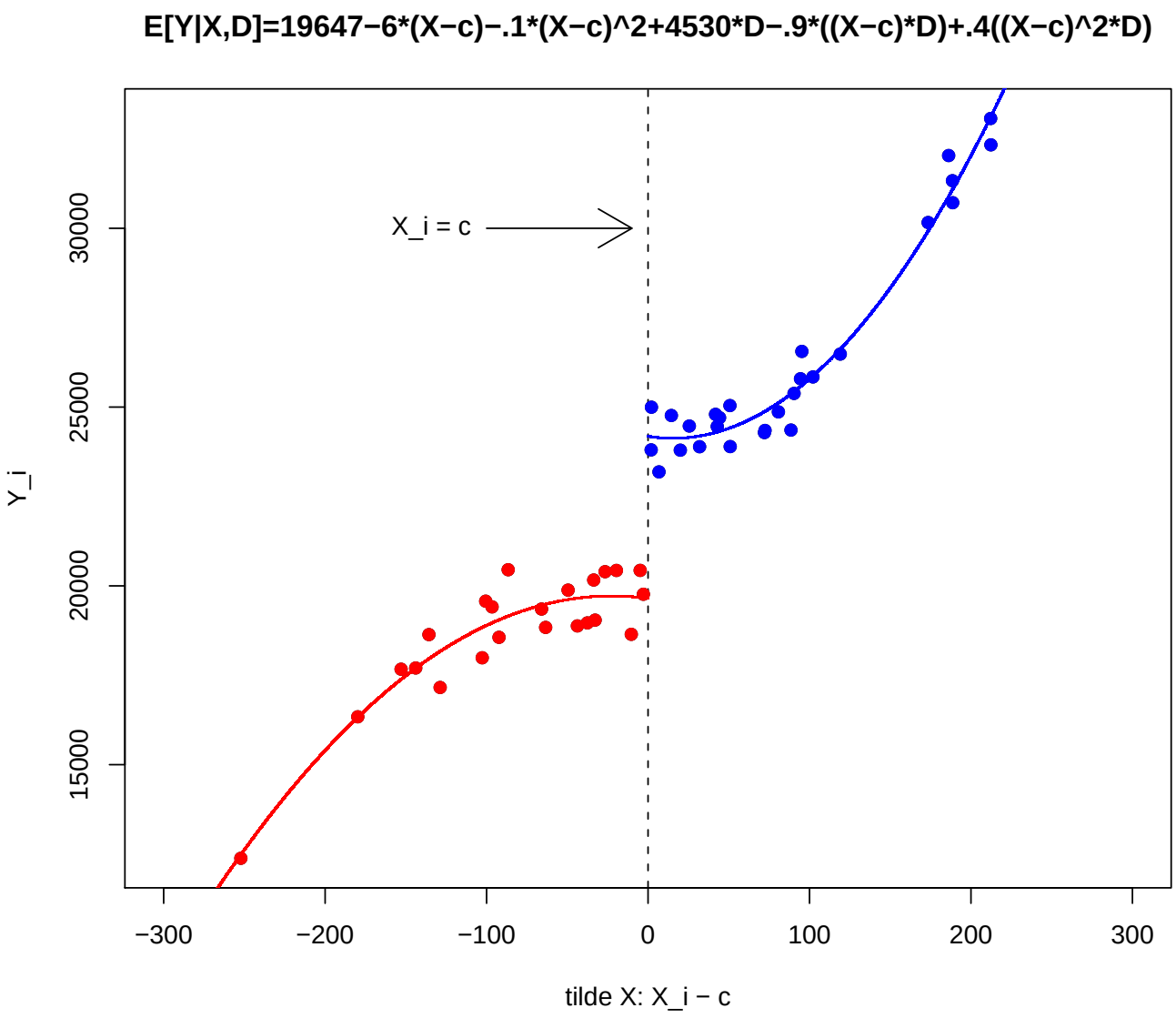
- Typically limit the data to some discontinuity window, $[c - h \leq X \leq c + h]$ first.
- Here we have $\alpha_0 = \mathbb{E}[Y_1 - Y_0|X = c]$ estimates effect at $X = c$

IMHO: This is good for just seeing your data. Quadratic is a nice balance of simple and flexible. Cubic too risky.

SRD: Linear with Differential Slope



SRD: Non-Linear Case



Option 2. Locally-weighted regression

More broadly, Non-parametric smoothers/ locally-weighted regressions

- just model on each side then take difference at $X = c$
- replaces worry of specification and discontinuity sample with decisions about smoothness of estimator (i.e. kernel bandwidth)

Automated selection procedures: choose something reasonable via cross-validation, with minimal user input:

- Imbens-Kalyanaraman (2012): See `IKbandwidth` in `RDD` package
- Calonico, Cattaneo, and Titiunik for automatic bandwidth selection for local polynomial models (see `rdrobust`). Now standard.
- One approach: do some simple models (like above) for yourself, use `rdrobust` and `RDD` to get estimates people can agree are reasonable (hopefully they agree!).
- However, increasing evidence that this approach requires very large samples and in practice produces undercoverage, high variance (and even wild) estimates, fails to run, or ends up including data farther from cutoff than you'd like.

Option 3. Gaussian process (GP) approach

First a detour to explain the GP.

If I tell you Y^* has mean 0 and variance σ^2, all you can believe is.

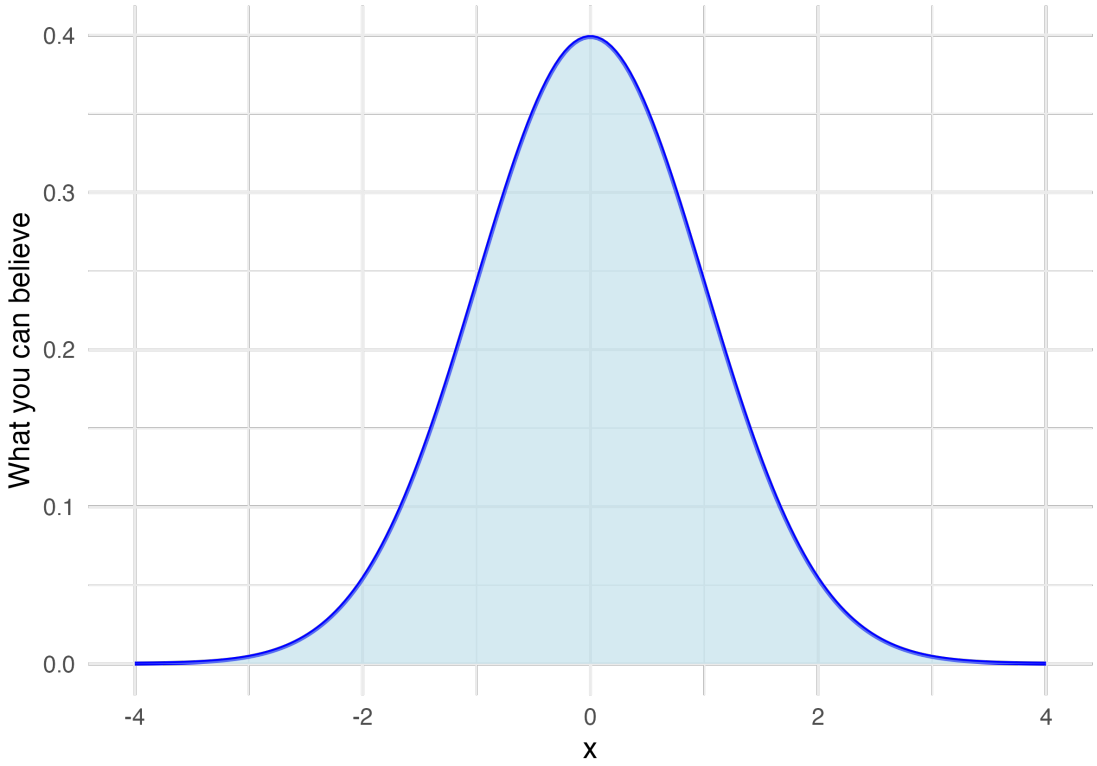


Figure: Total ignorance

GP intuition: Ignorance and neighbors

Now I tell you we see $Y_{obs} = -1$, and $cor(Y^*, Y_{obs}) = .8$. Then you believe:

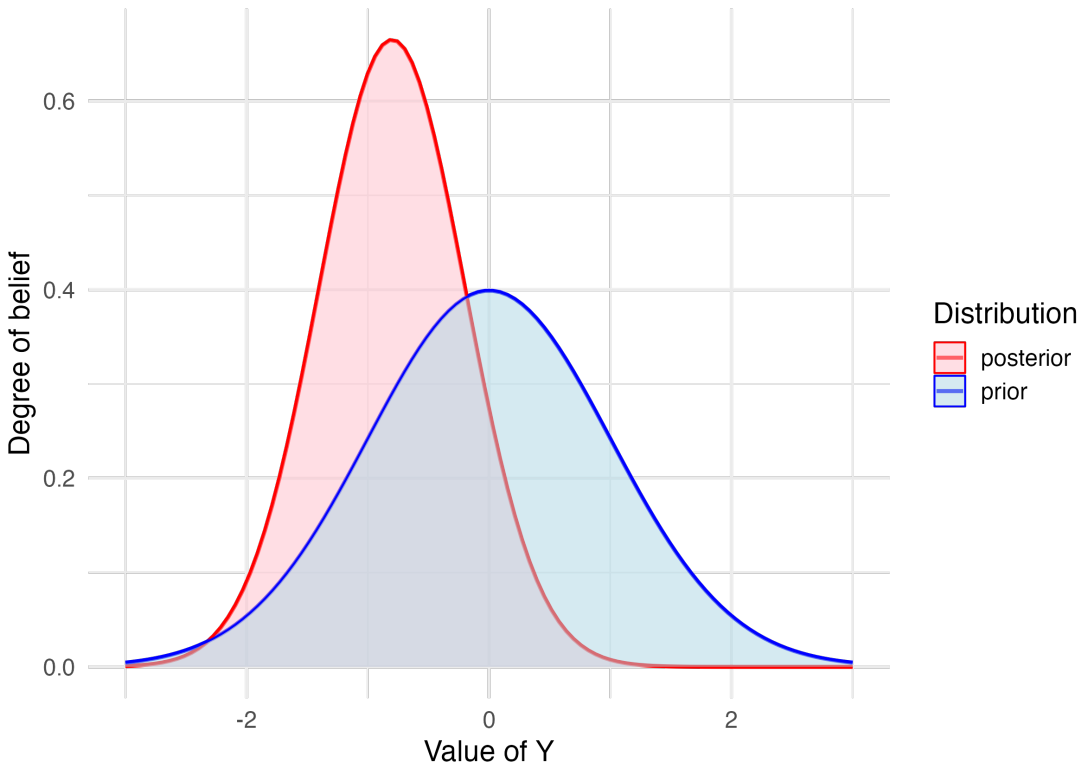


Figure: Informed ignorance

GP intuition: Ignorance and neighbors

Now suppose you have numerous observations of $\{X_i, Y_i\}_i^N$.

Big idea: observations nearer each other in X should have similar Y, whatever the values:

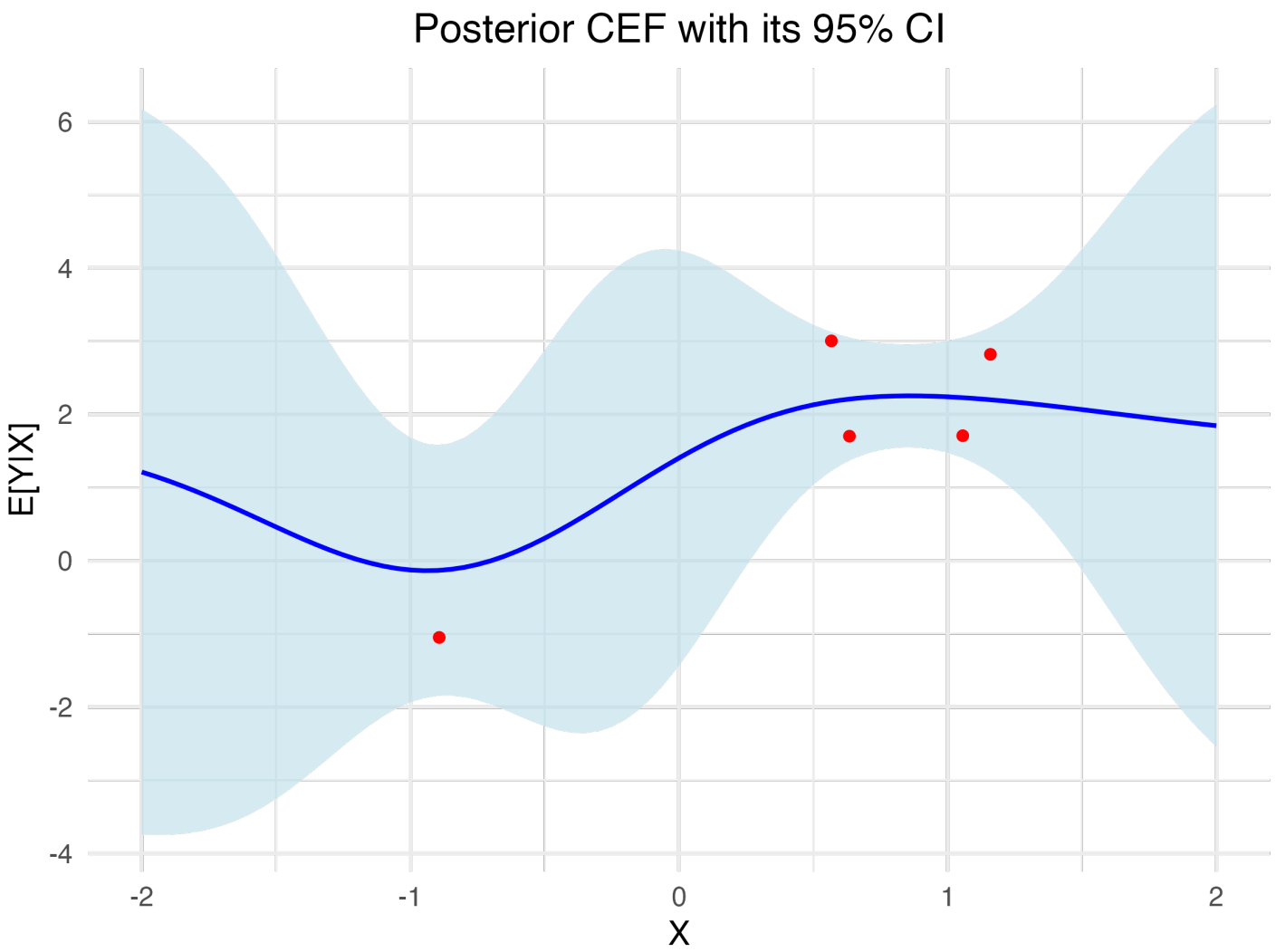
- That is, $cov(Y_i, Y_j)$ larger when X_i nearer X_j.
- E.g., $cov(Y_i, Y_j) = k(X_i, X_j) = e^{-\frac{||X_i - X_j||^2}{b}}$

At values of x^* near to observed points, the high covariance with observed Ys improves our guess about Y^*.

At values of x^* far from observed points, low covariance of observed outcomes with Y^* leaves us more uncertain about $p(Y^*)$.

Inferences given a few observations

Given GP assumptions and 5 observations of $\{X_i, Y_i\}$, ask for predictions at any X:



Written out,

1. $Y \sim \mathcal{N}(\mu, \Sigma)$, deferring how μ and Σ will be determined.
2. Choose a kernel function $k(\cdot, \cdot)$, with $cov(Y_i, Y_j) = \sigma_y k(X_i, X_j)$. This can be given a particular form, such as $k(X_i, X_j) = exp(-||X_i - X_j||^2/b)$.
3. Consider the kernel matrix $\mathbf{K}$, containing all the pairwise kernel evaluations, i.e. $\mathbf{K}_{i,j} = k(X_i, X_j)$.
4. Then,
$$Y|X \sim \mathcal{N}(\mu, \sigma_y \mathbf{K})$$
5. That model *goes through every point*, but we expect noise:
$$Y|X \sim \mathcal{N}(\mu, \sigma_y(\mathbf{K} + \sigma^2 I))$$
6. After centering and scaling Y we can use $\mu = 0$ and $\sigma_y = 1$,
$$Y|X \sim \mathcal{N}(0, \mathbf{K} + \sigma^2 I)$$
7. "Conditioning on the observations", beliefs about the distribution of test points (Y^*) is
$$Y^* \sim \mathcal{N}(\mathbf{K}_* \underbrace{(\mathbf{K} + \sigma^2 I)^{-1} Y}_{\text{coefs.}}, \underbrace{\mathbf{K}_{*,*} + \sigma^2 I}_{\text{prior var}} - \underbrace{\mathbf{K}_*^\top (\mathbf{K} + \sigma^2 I)^{-1} \mathbf{K}_*}_{\text{reduction due to data}}) \quad (1)$$

Case 3: Regression Discontinuity

RD requires estimating to the edge of the data, or slightly past it:

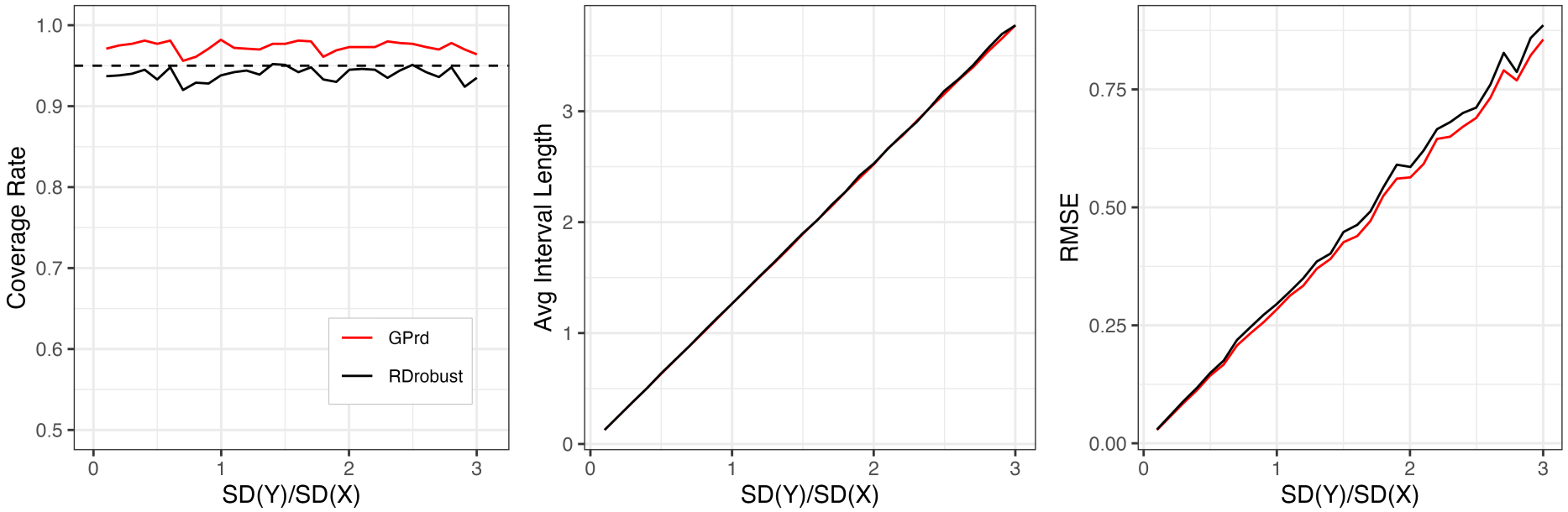
- $\widehat{Y(0)}_{Z=c}$: Train $\mathbb{E}[Y(0)|Z]$ on $Z < c$ and predict at $Z = c$
- $\widehat{Y(1)}_{Z=c}$: Train $\mathbb{E}[Y(1)|Z]$ on $Z > c$ and predict at $Z = c$

$$\hat{\tau} = \widehat{Y(1)}_{Z=c} - \widehat{Y(0)}_{Z=c}$$
$$\widehat{Var}(\hat{\tau}) = \widehat{Var}(\widehat{Y(1)}_{Z=c}) + \widehat{Var}(\widehat{Y(0)}_{Z=c})$$

- Easy to implement: In `GPss` package for `R`; http://doeunkim.org/gpss.
- Does not require large samples for reasoned inference

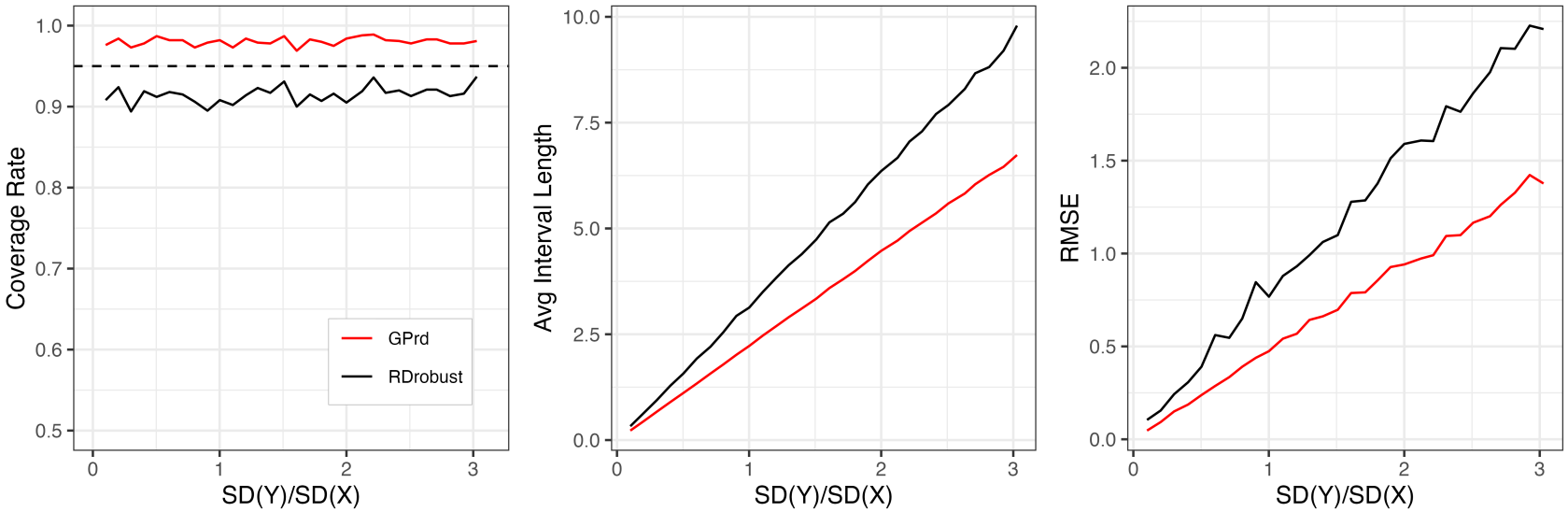
RD: random Z and Y, large sample

$\tau = 3$ (or $\tau = 0$), $N = 500$, `rdrobust` at defaults (robust estimate), `GPrd` at defaults



RD: random Z and Y, small sample

$\tau = 3$ (or $\tau = 0$), $N = 100$, `rdrobust` at defaults (robust estimate), `GPrd` at defaults



Turns out the improved RMSE is the big story, and relates to GP's regularization on each side whereas polynomials can go a little crazy.

Placebo cutoffs in close elections (application)

Lee (2004) close election setting:

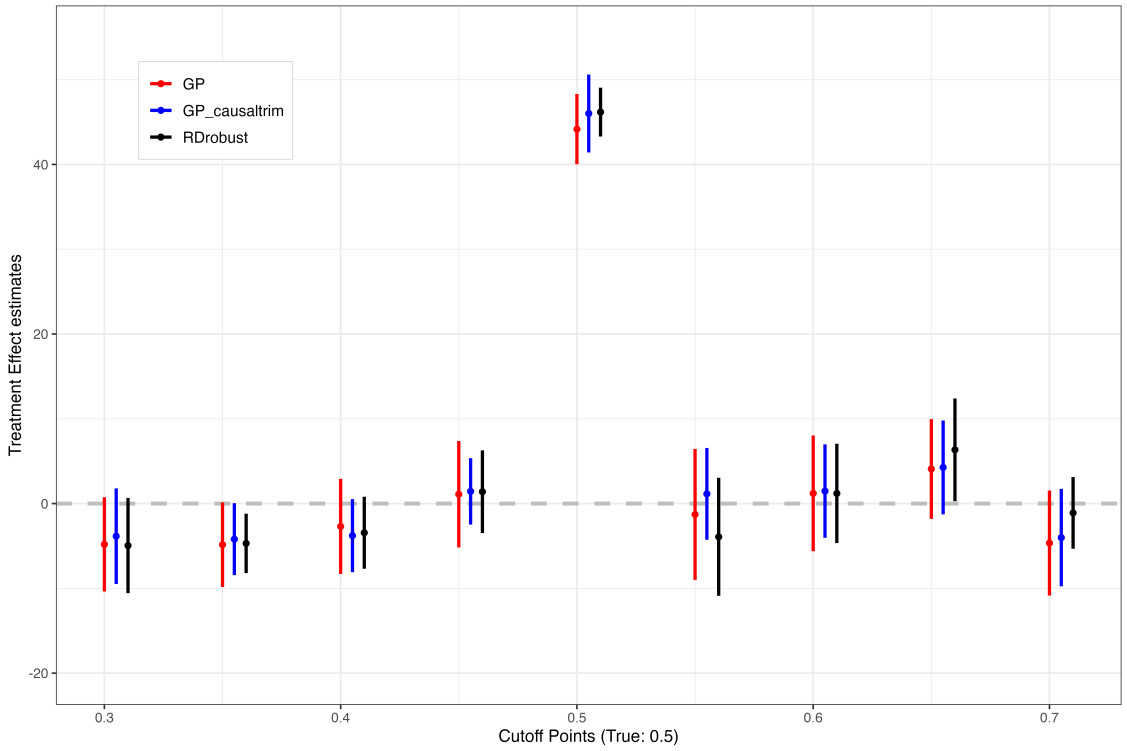
- Forcing variable: Democrats' vote share in US House election (1948 - 1990)
- Outcome variable: ADA score (liberal vote score of each representative)

Another twist: Should our RD estimates be informed by some *causal assumptions* regarding comparability?

- Should we worry if an automated RD procedure chooses a very wide bandwidth?
- In the `gp_causaltrim` approach we:
  1. use small bandwidth to limit comparison range: e.g. within 2% vote share, cov=cor=0.9 $\rightarrow$ b=0.005
  2. trim: remove data outside useful range ($\pm$10%, so scaling and σ^2 choice are focused)

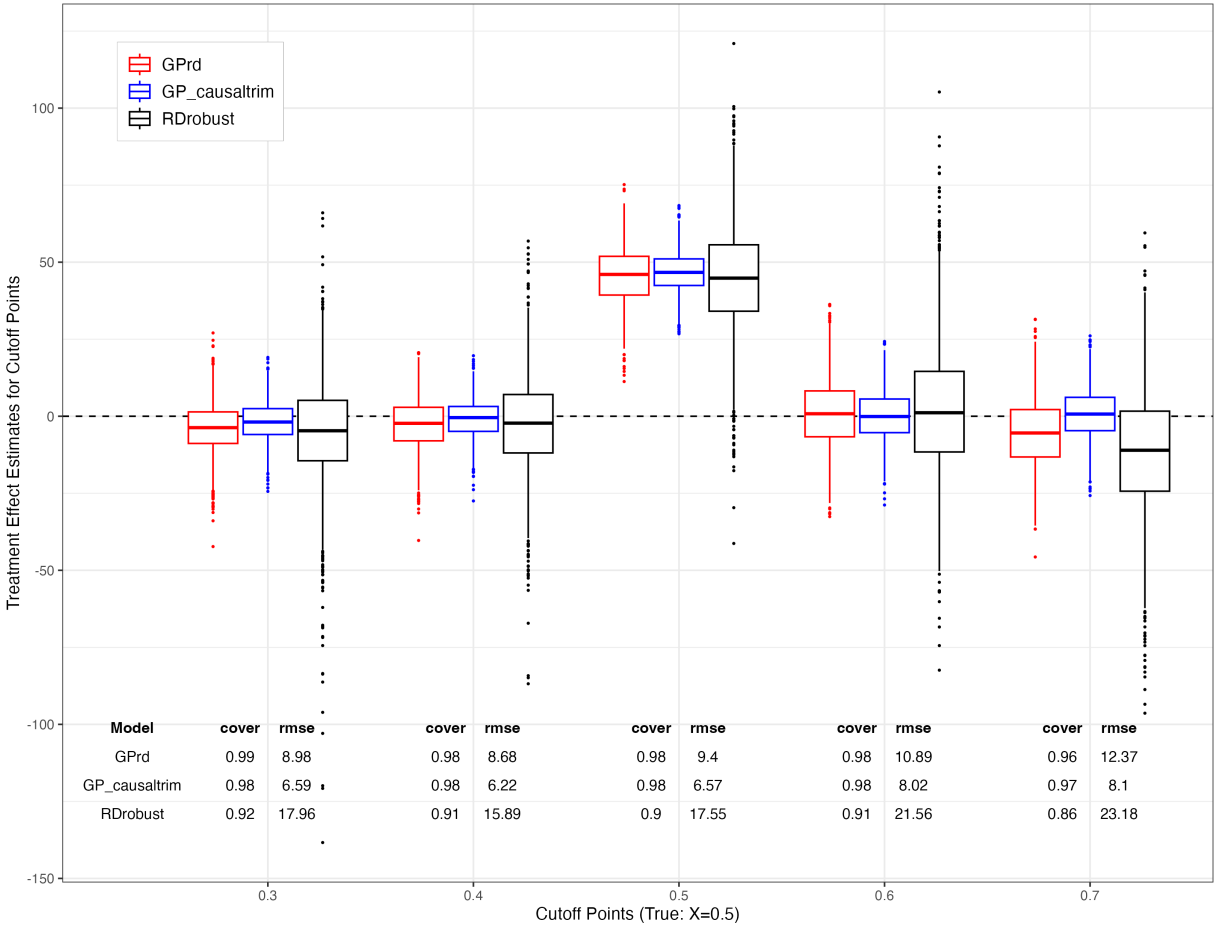
Placebo cutoffs in close elections: Large sample

Full sample: $N = 13,577$ (each analysis sees only part of this)



Placebo cutoffs in close elections: Small samples

Repeated sampling of sub-samples: $N = 200$ in each segment

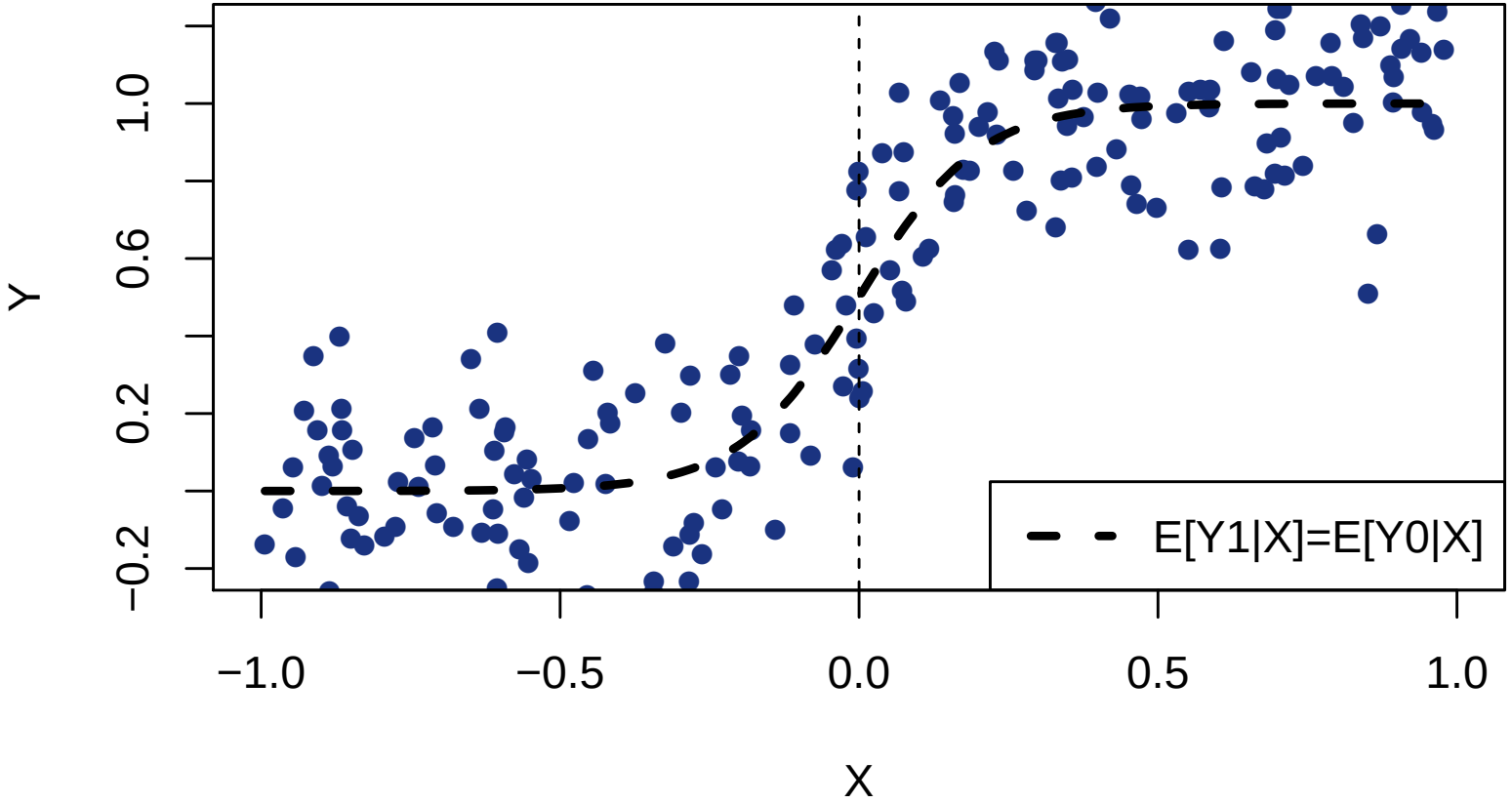


Model	cover	rmse	cover	rmse	cover	rmse	cover	rmse	cover	rmse
GPrd	0.99	8.98	0.98	8.68	0.98	9.4	0.98	10.89	0.96	12.37
GP_causaltrim	0.98	6.59	0.98	6.22	0.98	6.57	0.98	8.02	0.97	8.1
RDrobust	0.92	17.96	0.91	15.89	0.9	17.55	0.91	21.56	0.86	23.18

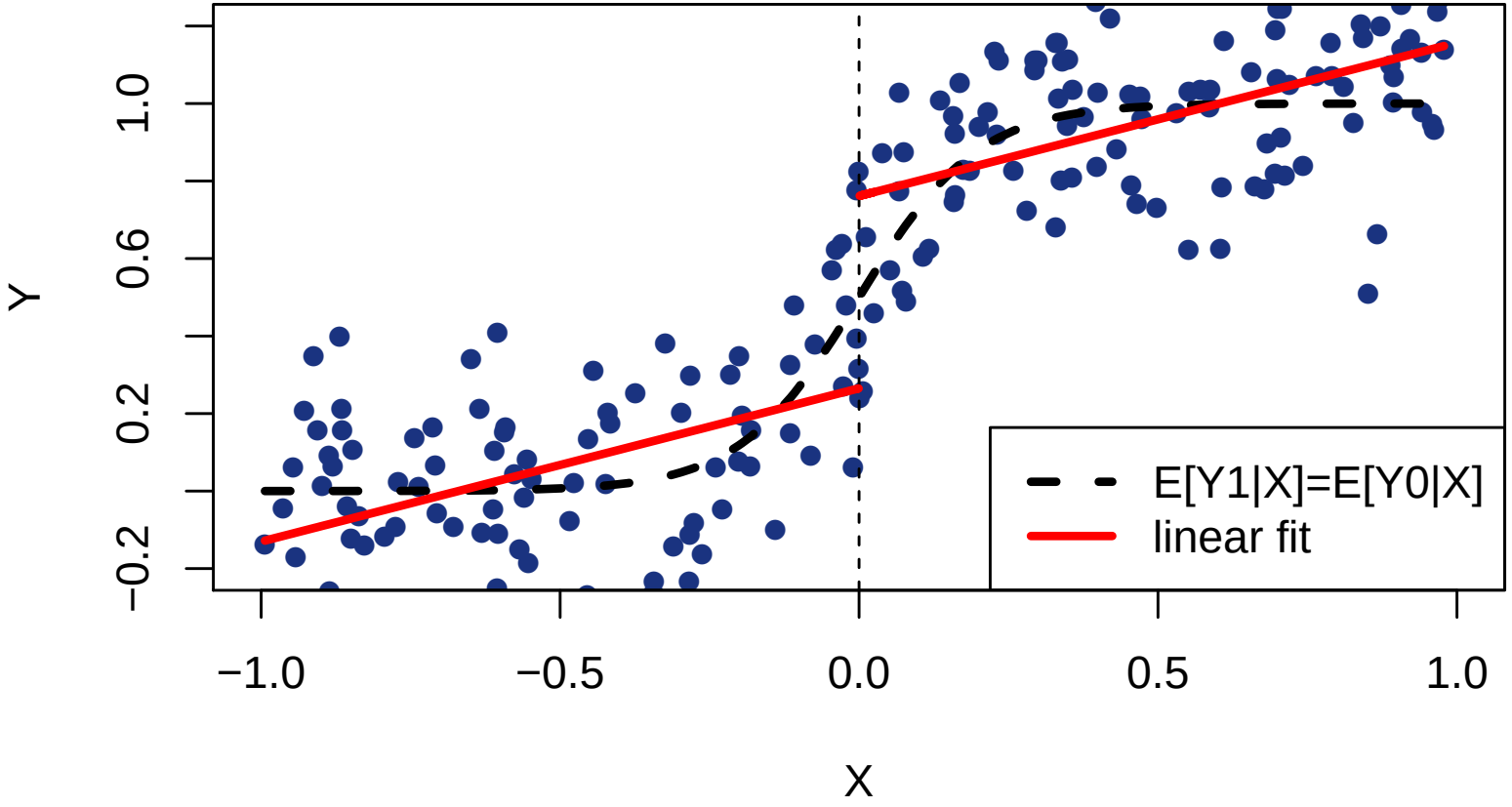
Sharp RDD: Falsification Checks

1. Specification sensitivity: Are results sensitive to alternative specifications?
2. Balance Checks: Do covariates Z jump at the threshold?
3. Check if jumps occur at placebo thresholds $c^{\star}$?
4. Sorting: Do units sort around the threshold? (density test)

Sharp RDD: Risk of Mispecification

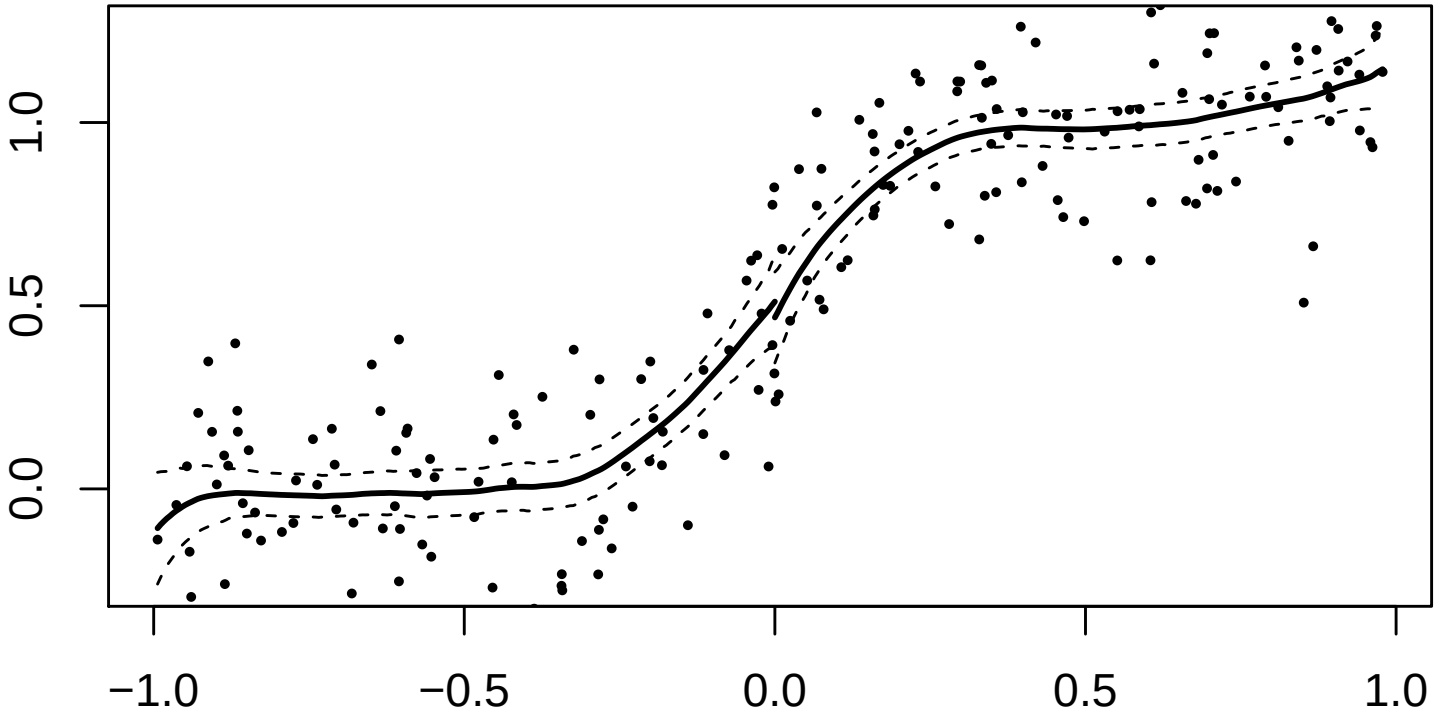


Sharp RDD: Risk of Mispecification



Sharp RDD: Risk of Mispecification

Using Imbens-Kalyanaraman bandwidth in `RDD`



SRDD: Balance Test

Test for comparability of agents around the cut-off:

- For potential outcomes to be continuous, any influencer (observed or unobserved) must also be continuous
- Consider pre-treatment covariate (or placebo outcome) Z_i
- Run the RDD regression using Z_i as the outcome. Use same estimation as for the real outcome. E.g.,

$$\mathbb{E}[Z_i|X_i, D_i] = \beta_0 + \beta_1(X_i - c) + \alpha_z D_i + \beta_3(X_i - c) \cdot D_i$$

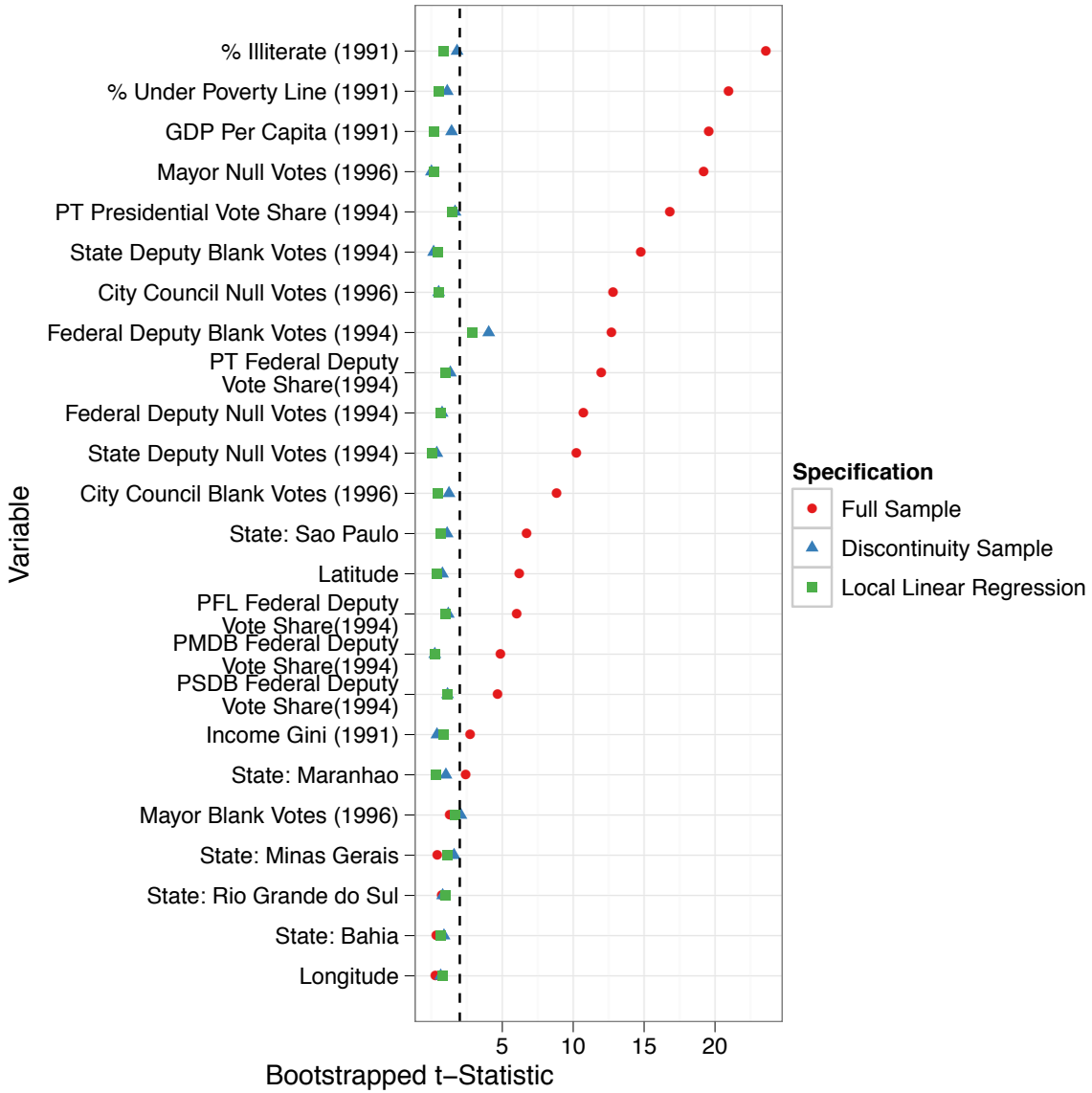
  should yield $\alpha_z = 0$ if Z is balanced at the threshold.

Finding a discontinuity in Z poses a challenge to the identification strategy.

- if you see imbalance at threshold on a post-treatment variable, is that a problem?

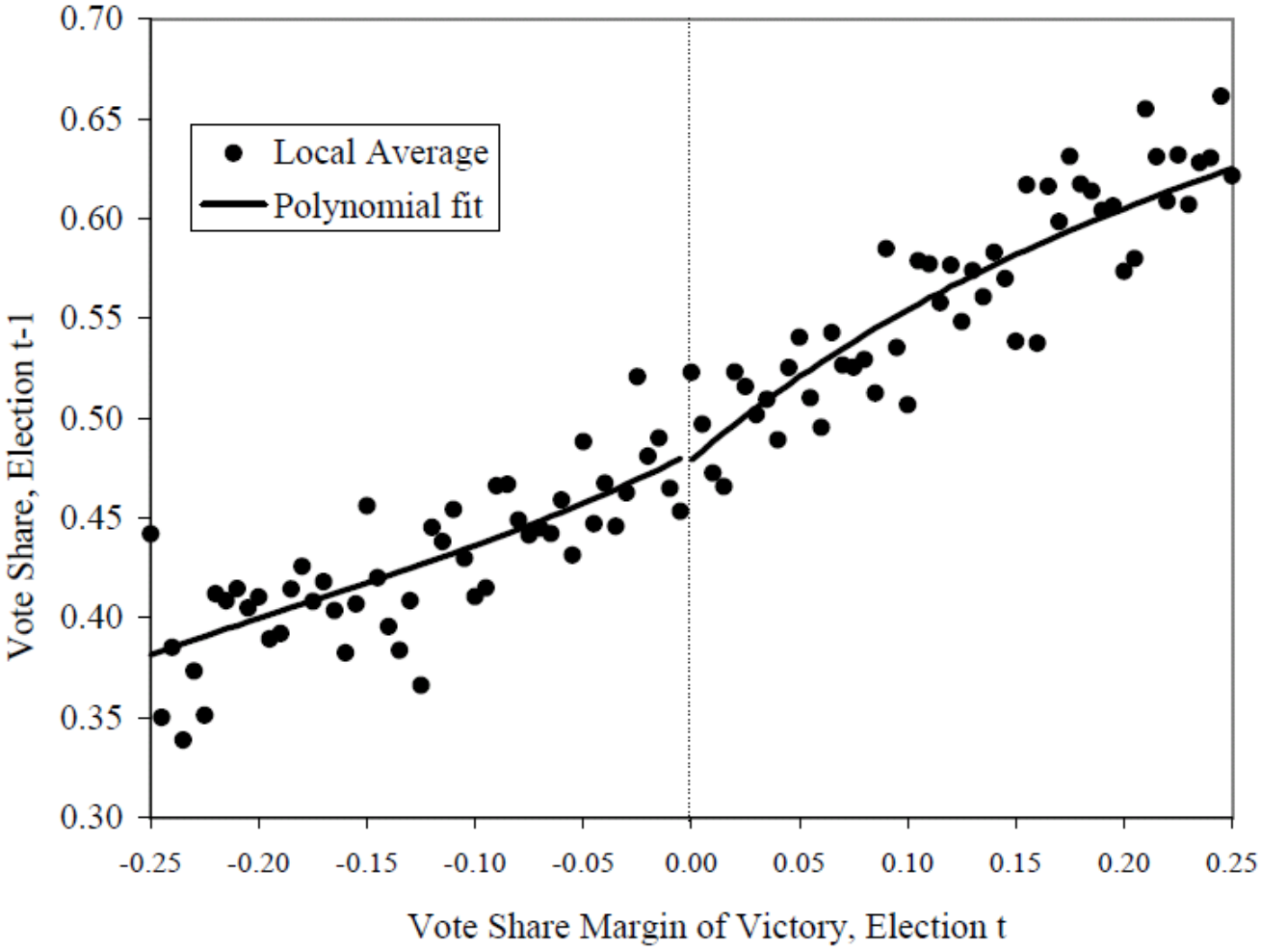
As usual, balance checks are falsification tests. Remember unobservables.

SRDD: Falsification Test



SRDD: Falsification Test

Especially convincing to do this with placebo outcome, i.e. covariates likely related to potential outcomes but not influenced by treatment.



SRDD: Placebo Thresholds

Let $c^\star$ be a placebo threshold value. Run models

$$\mathbb{E}[Y_i|X_i, D_i] = \beta_0 + \beta_1(X_i - c^\star) + \alpha_{fake} D_i + \beta_3(X_i - c^\star) \cdot D_i$$

and check if α_{fake} is large and significant.

- Usually we run separate models below and above c, with bandwidth to exclude c, to avoid misspecification due to inclusion of the actual jump.
- Detecting a jump at placebo threshold requires an explanation.
  - e.g. if other treatments occurred near $c^\star$, it could be okay
  - but problematic if it implies unknown treatments or unobserved influences may account for effect at c
  - can also reveal specification problems, as your model could be guessing a discontinuous jump occurs where it was actually smooth

SRDD: Sorting

Can subjects self-sort around threshold?

- Can they exercise fine control over the running variable?
- Can administrators strategically choose what is used as running variable or as threshold?
- Either can allow sorting of agents around the cut-off, s.t. those below may differ on average form those just above
- Does not necessarily invalidate the design unless sorting is (a) precise, and (b) widespread
  - (a) imprecise sorting is okay as long as you can find narrow enough bandwidth around $X = c$ s.t. those to left and right of c are no different
  - (b) if some people can sort very precisely, RDD may be biased but less biased then naive comparison

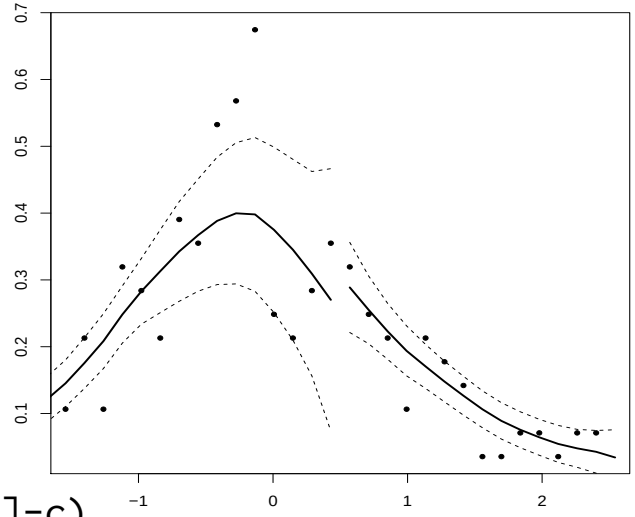
SRDD: Sorting Around the Threshold

Test for discontinuity in density of forcing variable:

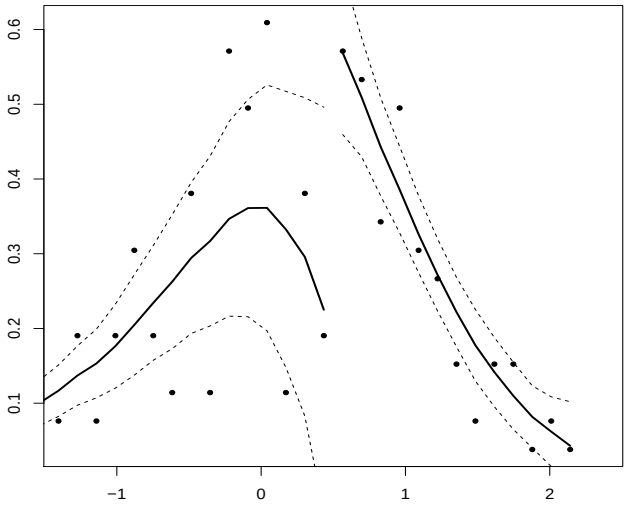
- Visual Histogram Inspection:
  - Construct equal-sized non-overlapping bins of the forcing variable s.t. no bin includes points both above and below c
  - For each bin, compute the number of observations and plot the bins to see if there is a discontinuity at the cut-off
- Formal test: McCrary (2008)

Self-sorting simulation

```
>N=200
>runvar=rnorm(N, mean = 0, sd=1)
>c=.5
>DCdensity(runvar, cutpoint=c)
[1] 0.5857803
```




```
> sortprob=rep(0,N)
> sortprob[runvar<c]=.25*exp(runvar[runvar<c]-c)
> selfsorts=rbinom(N, size=1, prob = sortprob)
> runvar[selfsorts==1]=.5+
runif(sum(selfsorts),.001,.5)
> DCdensity(runvar, cutpoint=c)
[1] 0.0001952333
```



Bundled (or just other) Treatments

- The achilles heal of RD identification: other treatments at the same cutoff.
  - violates continuity assumpion.
  - essentially a confounder
- Ask yourself: what else changes at c?
- Worry about commonly used eligibility criteria, political borders.

Fuzzy Regression Discontinuity Design

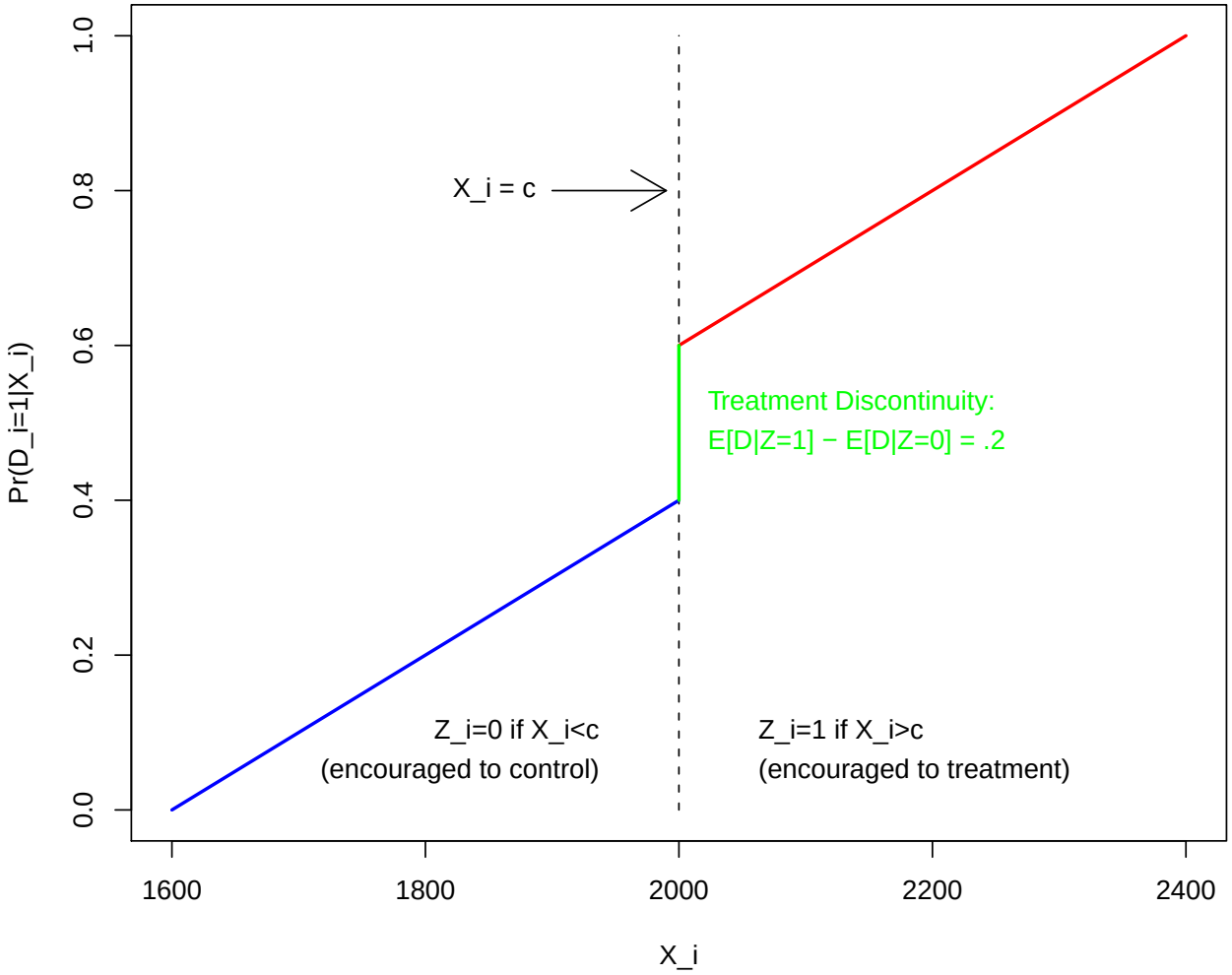
- Threshold may not perfectly determine treatment exposure, but it creates a discontinuity in the probability of treatment exposure
- Incentives to participate in a program may change discontinuously at a threshold, but the incentives are not powerful enough to move all units from non-participation to participation
- Should remind you of IV!

We can use discontinuities to produce instrumental variable estimators of the effect of the treatment (close to the discontinuity)

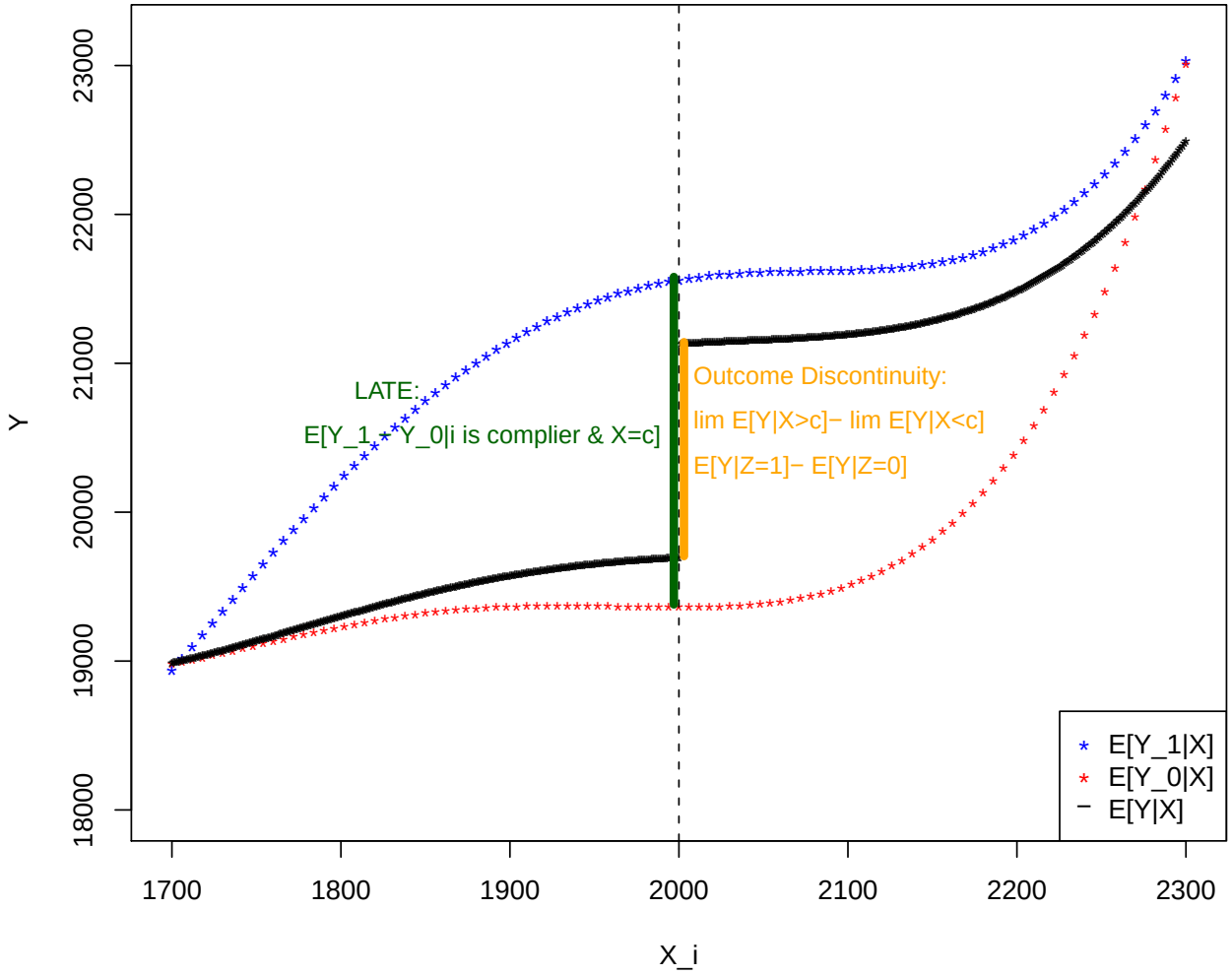
Fuzzy RDD Example:

- Probability of being offered a scholarship may jump at a certain SAT score threshold (when applicants are given "special consideration")
- For applicants with scores very close to the threshold:
  - scholarship eligibility should be random (so $Z_i \perp\!\!\!\perp D_{1i}, D_{0i}$)
  - we may think tiny change in score, being nearly random, only effects outcome through scholarship chances ($Z_i \perp\!\!\!\perp Y_{0i}, Y_{1i}$)
  - If both, we have a good instrument.
- Who are compliers in this framework?

Fuzzy RDD: Discontinuity in $\mathbb{E}[D|X]$



Fuzzy RDD: Discontinuity in $\mathbb{E}[Y|X]$



Fuzzy RDD: Identification

Identification Assumption

For sufficiently small ϵ s.t. $X_i - \epsilon < c < X_i + \epsilon$, we have usual IV assumption:

1. *Ignorability (and Exclusion):* $\{Y_{1i}, Y_{0i}, D_{1i}, D_{0i}\} \perp\!\!\!\perp Z_i$
2. *Relevance:* $\mathbb{E}[D_{1i}] \neq \mathbb{E}[D_{0i}]$
3. *Monotonicity:* $D_{1i} \geq D_{0i}\ \forall i$

Identification Result

$$\begin{aligned}
\alpha_{FRDD} &= \mathbb{E}[Y_{1i} - Y_{0i} | X_i = c \text{ and } i \text{ is a complier}] \\
&= \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = c] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = c]}{\lim_{x \downarrow c} \mathbb{E}[D_i | X_i = c] - \lim_{x \uparrow c} \mathbb{E}[D_i | X_i = c]} \\
&= \frac{\textit{outcome discontinuity}}{\textit{treatment discontinuity}} \\
&\approx \frac{\hat{\mathbb{E}}[Y | Z = 1, X = c] - \hat{\mathbb{E}}[Y | Z = 0, X = c]}{\hat{\mathbb{E}}[D | Z = 1, X = c] - \hat{\mathbb{E}}[D | Z = 0, X = c]}
\end{aligned}$$

Fuzzy RDD: Estimation

- Cut the sample to a small window above and below the threshold (discontinuity sample)
- Code instrument $Z_i = \mathbb{1}_{\{X>c\}}$
- Fit 2SLS/IV: $Y_i = \beta_0 + \beta_1(X_i - c) + \beta_2 Z_i(X_i - c) + \alpha D_i$ where D_i is instrumented with Z_i
- Better: use polynomials
- Better still: estimate the outcome discontinuity and treatment discontinuity, then divide. Can use any estimator this way. `rdrobust` can do this for you.

Early Release Program (HDC)

- Prison system in many countries is faced with overcrowding and high recidivism rates after release.
- Early discharge of prisoners with electronic monitoring has become a popular policy
- Difficult to estimate impact of early release program on future criminal behaviour: best behaved inmates are usually the ones to be released early.
- Marie (2008) considers Home Detention Curfew (HDC) scheme in England and Wales:
  - Fuzzy RDD: Only offenders sentenced to more than 88 days in prison are eligible for HDC, but obviously, not all those with longer sentences are offered HDC
- How valid are the identification assumptions?

Table 1: Descriptive Statistics for Prisoners Released by Length of Sentence and HDC and Non HDC Discharges

Panel A - Released Before 3 Months:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	12.2	-	12.2
Mean Age	29.5	-	29.5
Percentage Incarcerated for Violence	17.6	-	17.6
Mean Number Previous Offences	8.8	-	8.8
Recidivism within 12 Months	52.4	-	52.4
Sample Size	42,987	0	42,987
Panel B - Released Between 3 and 6 Months:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	8.8	8.8	8.8
Mean Age at Release	27.6	30.8	28.4
Percentage Incarcerated for Violence	20.3	18.3	19.8
Mean Number Previous Offences	10	6.5	9.1
Recidivism within 12 Months	60	30.2	52.6
Sample Size	52,091	17,222	69,313

Table 2: Descriptive Statistics for Prisoners Released by Length of Sentence and HDC and Non HDC Discharges and +/-7 Days Around Discontinuity Threshold

Panel A - Released +/- 7 Days of 3 Months (88 Days) Cut-off:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	10.5	9.7	10.3
Mean Age at Release	28.9	30.7	29.3
Percentage Incarcerated for Violence	19.8	18.2	19.4
Mean Number Previous Offences	9.5	5.7	8.7
Recidivism within 12 Months	54.6	28.1	48.8
Sample Size	18,928	5,351	24,279
Panel B - Released +/- 7 Days of 3 Months (88 Days) Cu-off:			
Day of Release around Cut-off	**- 7 Days**	**+ 7 Days**	**Total**
Percentage Female	11	10.2	10.3
Mean Age at Release	28.8	29.4	29.3
Percentage Incarcerated for Violence	17.1	19.7	19.4
Mean Number Previous Offences	9.1	8.6	8.7
Recidivism within 12 Months	56.8	47.9	48.8
Percentage Released on HDC	0	24.4	22
Sample Size	2,333	21,946	24,279

Figure 1: Proportion Discharged on HDC by Sentence Length

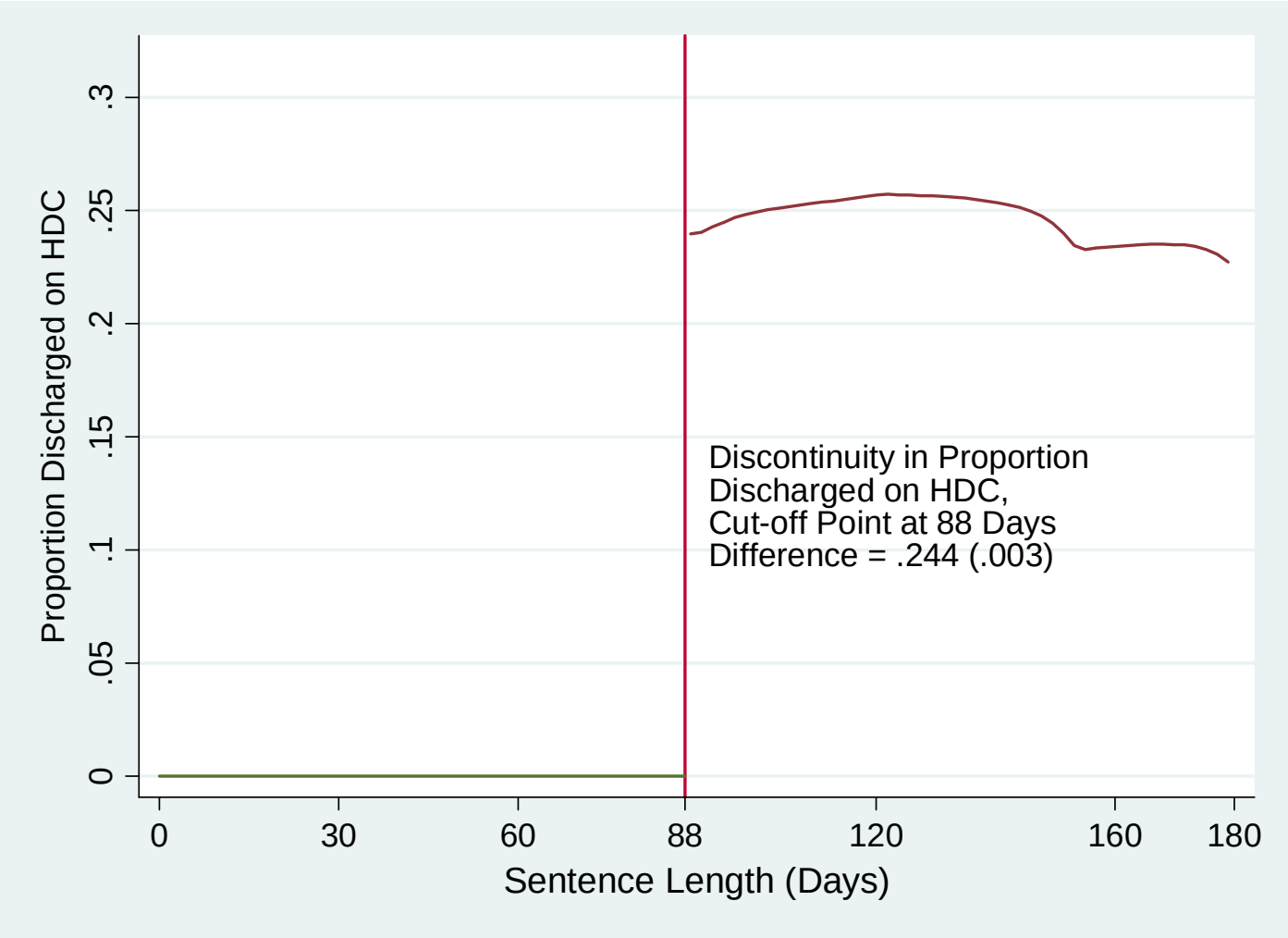


Figure 2: Mean Number of Previous Offence by Sentence Length

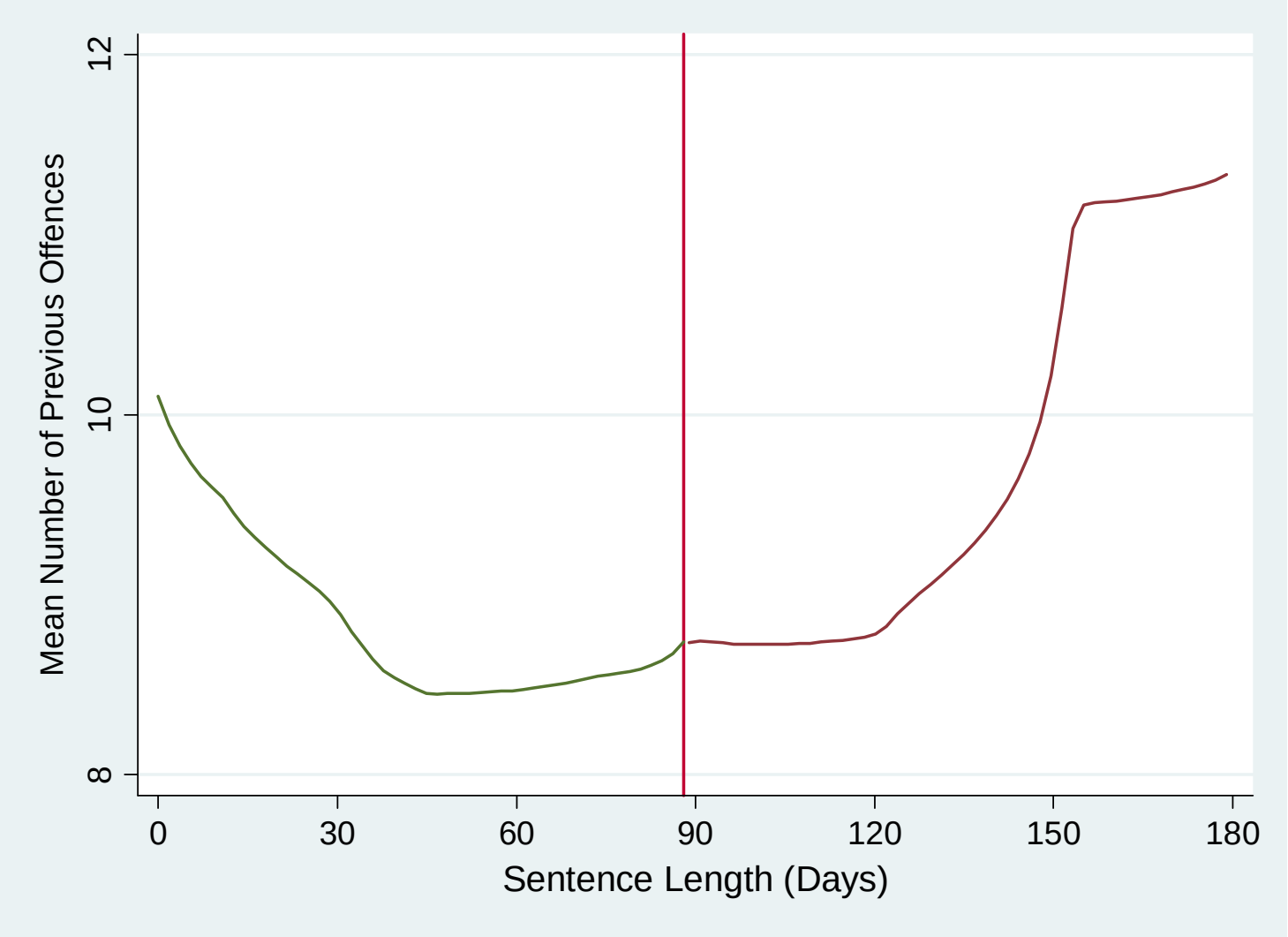


Table 4: RDD Estimates of HDC Impact on Recidivism – Around Threshold

	Dependent Variable = Recidivism Within 12 Months Estimation on Individuals Discharged +/- 7 Days of 88 Days Threshold		
	(1)	(2)	(3)
Estimated Discontinuity of HDC Participation at Threshold ($HDC^{+} - HDC^{-}$)	.243 (.009)	.223 (.009)	.243 (.003)
Estimated Difference in Recidivism Around Threshold ($Rec^{+} - Rec^{-}$)	-.089 (.011)	-.059 (.009)	-.044 (.014)
Estimated Effect of HDC on Recidivism Participation ($Rec^{+} - Rec^{-}$)/ ($HDC^{+} - HDC^{-}$)	-.366 (.044)	-.268 (.044)	-.181 (n.a.)
Controls	No	Yes	No
PSM	No	No	Yes
Sample Size	24,279	24,279	24,279

For what group is this the ATE?

Internal and External Validity

Internal validity:

- Because the continuity of POs is sometimes reasonably defensible, RDD tends to be more credible
- Even if imperfect it is likely to be bias reducing
- Main threats tend to be other treatments at same threshold, and specification problems.

External validity:

- You get the average effect of the sub-population with X_i close to c
- Fuzzy RDD further restricts this to the compliers with X_i close to c
- Only with strong assumptions (e.g., homogenous treatment effects) can we estimate the overall average treatment effect